Worldgen comparison: Vanilla, Paper and Horizon, third round

27 October 2026. Minecraft 26.3. Seeds 12345, 1 and 8675309. 1,024 x 1,024 blocks around 0,0 (4,096 chunks) in the Overworld, the Nether and the End; the animal and speed tests as described below.

Question

Can Horizon replace our Paper + Chunky seed generator for vanilla worlds? This round fixes what the second report left open, looks at animals and entity save fields for the first time, and measures speed with Horizon's own pre-generation command.

Setup

- **Vanilla**: the 26.3 server, chunks force-loaded in batches, ticks frozen, game rules set in every dimension. A **second vanilla run** of each seed shows how much vanilla differs from itself.
- **Paper + Chunky**: Paper 26.3 build 223 with Chunky, as our seed generator runs.
- **Horizon 0.2.69**: the current upstream release.
- **Horizon with our fixes**: upstream main of 27 October, which now holds 20 of our fixes, plus the open ones: the maintainer's #2698 (carrying our #2544 and #2586), #2608, #2706 and #2716. This is the test build royalpixels-2026-10-27.
- **Animals**: our seed generator runs with vanilla's default game rules, so chunk generation places animals. This test generates 512 x 512 blocks of the Overworld with mob spawning on and ticks frozen, and compares every animal by the chunk it stands in.
- **Speed**: Horizon generates with its own `/pregen` command, the counterpart of Chunky, on the same machine (12 threads) and the same area as Paper.

Summary

	Vanilla (2nd run)	Paper + Chunky	Horizon 0.2.69	Horizon with our fixes
Blocks different, Overworld	0.17 %	0.34 %	0.38 %	0.15 %
Blocks different, Nether	0.32 %	0.65 %	0.76 %	0.23 %
Blocks different, End	0	24	0	0
Structure starts equal (Overworld + Nether)	426 of 427	421 of 427	0 of 427	418 of 427
Same loot as vanilla	-	99.1 %	86.4 %	99.4 %
Points of interest (vanilla: 288)	288	287	0	292
End entities (vanilla: 30)	30	30	0	30
Nether heightmap columns different	0	0	12,288	0
Generation time / CPU / peak memory	6.1 min / 14.1 min / 6.9 GB	2.1 min / 7.6 min / 7.3 GB	2.0 min / 6.7 min / 1.3 GB	2.5 min / 9.0 min / 1.3 GB

Horizon before and after

	RoyalPixels build of 26 Oct (first report)	With this round of fixes
Blocks different, Overworld	0.40 %	0.15 %
Blocks different, Nether	0.76 %	0.23 %
Blocks different, End	188	0
Same loot as vanilla	95.3 %	99.4 %
Structure starts equal	417 of 427	418 of 427
Block entities missing, Overworld	3,313	2,031

Animals

Seed	Vanilla animals	Horizon: same position / total	Second vanilla run: same position / total	Chunks with other animals: Horizon / second vanilla run
12345	25	21 / 23	25 / 25	2 / 0
1	17	13 / 17	12 / 16	2 / 3
8675309	140	117 / 142	112 / 146	14 / 15

Before this round, a Horizon world generated with ticks frozen had **no animals at all** (seed 12345: 0 of 25). Now Horizon places them as vanilla does. The chunks where animals still differ are chunks where vanilla's own decoration differs between runs (a giant mushroom, a tree), so a second vanilla run differs there too. In seed 12345 the two such chunks lie next to the spawn, where the startup order now follows vanilla's.

Speed

Seed	Paper + Chunky: wall / CPU seconds	Horizon /pregen: wall / CPU seconds
12345	45 / 163	27 / 167
1	38 / 139	24 / 143
8675309	42 / 156	28 / 157
Total	125 s	79 s (1.6x faster)

On the small 1,024-block area, Horizon is 1.6 times as fast as Paper + Chunky with a seventh of the memory (1.1 against 7.3 GB). On a larger area the start-up share falls away: a 2,000 x 2,000 Overworld (15,876 chunks) took Horizon 36 seconds, 443 chunks per second, about 2.8 times Paper's rate. Two thirds of Horizon's time goes into the raw terrain (noise and surface rules), a fifth into decoration.

Result

- **Horizon with our fixes matches vanilla as closely as a second vanilla run, or closer.** Overworld blocks differ 0.15 % (second vanilla run 0.17 %), Nether 0.23 % (0.32 %), the End not at all. Loot 99.4 % (Paper 99.1 %).
- **Animals and entities** now match vanilla within vanilla's own run-to-run differences: positions, variants, sound variants, babies, villagers, cats, golems, horses, foxes, rabbits, goats and minecarts.
- **Speed**: faster than Paper + Chunky on every seed, 2.8 times on a large area.
- **Horizon 0.2.69 as released still cannot be used**: no structure starts, no points of interest, wrong Nether heightmaps, no End entities.

What this round fixed

- **Animals at generation** (#2586, carried in #2698): placed while generating, as vanilla, with its random; before, only on a game tick, so a frozen world had none.

- **Variants** (#2541, #2585, merged): pigs, cows, chickens, frogs, foxes and rabbits kept their variant and sound variant; goats their horns; horses a health above their maximum.
- **Structure mobs' save fields** (#2544, carried in #2698): villagers, cats, iron golems, drowned and jockeys saved as vanilla's proto chunk keeps them.
- **Minecarts** (#2543, merged): `HasTicked` and `Fire` as vanilla.
- **Bee nests**: generated nests were empty; fixed on main in the meantime (our #2542 was closed for that, its test is kept).
- **Unlit edge** (#2619, merged): the outermost ring of a pre-generated area was saved without light.
- **Spawn area of an empty server** (#2608): the dedicated server kept and generated an area around the spawn while nobody was on. It caused about 4,100 stone blocks and a mineshaft flag near the spawn to differ, and cost CPU when idle.
- **Speed** (#2706): the surface step skips writes that change nothing (Nether surface 16 % faster, same output).

Remaining differences

- **Structure starts**: 9 of 427 differ, all mineshaft and fortress flags in chunks at the edge of the area, where Paper differs too, and one ocean ruin that a second vanilla run also changes.
- **Points of interest**: bee nests sit on trees that vanilla itself places differently between runs (a second vanilla run differs by 14, Horizon by 16).
- **Block entities**: about 2,000 sculk block entities differ in each direction in the ancient city of seed 12345, where sculk spreads randomly; a second vanilla run differs the same way.
- **Light format**: Horizon also saves all-dark sky light arrays deep underground, where vanilla leaves them out. The light values are the same; the files are slightly larger.
- **Forced area**: vanilla makes the two rings around a force-loaded area full chunks too; Horizon keeps them as unfinished chunks. With Chunky-style pre-generation this does not matter.

All numbers

Overworld: compared with vanilla

	Vanilla (2nd run)	Paper + Chunky	Horizon 0.2.69	Horizon with our fixes
Blocks different	2,013,367 (0.17 %)	4,145,049 (0.34 %)	4,563,375 (0.38 %)	1,807,354 (0.15 %)
Chunks with a block difference	5,581 of 12,288	11,159 of 12,288	11,524 of 12,288	4,826 of 12,288
Biome cells different	0	521	0	0
Structure starts equal	116 of 117	112 of 117	0 of 117	109 of 117
Structure references different	0	0	0	0
Block entities different / missing / extra	22 / 2,208 / 2,112	969 / 3,175 / 3,197	1,225 / 4,236 / 75	137 / 2,031 / 1,865
Entities (vanilla / this)	395 / 393	395 / 377	395 / 384	395 / 382
Points of interest (vanilla / this)	288 / 288	288 / 287	288 / 0	288 / 292
Surface heightmap columns different	1,940	3,689	3,470	1,696

Nether: compared with vanilla

	Vanilla (2nd run)	Paper + Chunky	Horizon 0.2.69	Horizon with our fixes
Blocks different	2,539,519 (0.32 %)	5,212,373 (0.65 %)	6,098,199 (0.76 %)	1,814,351 (0.23 %)
Chunks with a block difference	4,597 of 12,288	8,355 of 12,288	7,997 of 12,288	3,926 of 12,288
Biome cells different	0	0	0	0
Structure starts equal	310 of 310	309 of 310	0 of 310	309 of 310
Structure references different	0	0	0	0
Block entities different / missing / extra	0 / 0 / 0	129 / 0 / 0	129 / 0 / 0	0 / 0 / 0
Entities (vanilla / this)	377 / 377	377 / 377	377 / 377	377 / 377
Points of interest (vanilla / this)	0 / 0	0 / 0	0 / 0	0 / 0
Surface heightmap columns different	0	0	12,288	0

End: compared with vanilla

	Vanilla (2nd run)	Paper + Chunky	Horizon 0.2.69	Horizon with our fixes
Blocks different	0 (0.00 %)	24 (0.00 %)	0 (0.00 %)	0 (0.00 %)
Chunks with a block difference	0 of 12,288	24 of 12,288	0 of 12,288	0 of 12,288
Biome cells different	0	0	0	0
Structure starts equal	0 of 0	0 of 0	0 of 0	0 of 0
Structure references different	0	0	0	0
Block entities different / missing / extra	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
Entities (vanilla / this)	30 / 30	30 / 30	30 / 0	30 / 30
Points of interest (vanilla / this)	0 / 0	0 / 0	0 / 0	0 / 0
Surface heightmap columns different	0	9	518	0

Loot: what a player finds when opening each container

	Paper + Chunky	Horizon 0.2.69	Horizon with our fixes
Containers with a vanilla loot table	2,003	2,003	2,003
Same loot as vanilla	1,985 (99.10 %)	1,731 (86.42 %)	1,990 (99.35 %)
Different loot	1	175	2
Container missing	17	97	11
Chest minecarts same / different / missing	244 / 0 / 24	13 / 220 / 35	238 / 0 / 30

Save format checks

	Vanilla	Vanilla (2nd run)	Paper + Chunky	Horizon 0.2.69	Horizon with our fixes
Nether/End chunks with the vanilla yPos 0	48,600 of 48600	48,600 of 48600	45,414 of 45414	0 of 27744	27,744 of 27744
Structure starts saved (Overworld + Nether)	476	476	433	0	427
Containers saved with their loot table	2,257	2,256	2,068	997	2,050
Containers saved with pre-rolled items	8	8	8	971	8
Trial spawners with an empty spawn_data	0	0	0	197	0
Block entities without an id	0	0	0	0	0
POI region files	21	21	18	0	12

Speed and load

	Vanilla	Vanilla (2nd run)	Paper + Chunky	Horizon 0.2.69	Horizon with our fixes
Generation time, 3 seeds x 3 dimensions	6.4 min	6.1 min	2.1 min	2.0 min	2.5 min
CPU time	13.7 min	14.1 min	7.6 min	6.7 min	9.0 min
Peak memory (RSS)	6.8 GB	6.9 GB	7.3 GB	1.3 GB	1.3 GB